Light

GCSE Computer Science — FDE Cycle and Systems (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the purpose of the fetch-decode-execute cycle?

- ○ **A.** To process instructions from a program one at a time in a continuous sequence
- ○ **B.** To download files from the internet
- ○ **C.** To display images on the monitor
- ○ **D.** To save files to the hard drive

Q2 (1 mark)

What happens during the 'fetch' stage of the FDE cycle?

- ○ **A.** The CPU performs a calculation
- ○ **B.** The next instruction is retrieved from memory
- ○ **C.** The instruction is interpreted to determine the operation
- ○ **D.** The result is stored in memory

Q3 (1 mark)

What happens during the 'decode' stage of the FDE cycle?

- ○ **A.** The instruction is retrieved from memory
- ○ **B.** The result of the instruction is stored
- ○ **C.** The Control Unit interprets the instruction to determine what action is required
- ○ **D.** The Program Counter is reset to zero

Q4 (1 mark)

What happens during the 'execute' stage of the FDE cycle?

- ○ **A.** The instruction is deleted from memory
- ○ **B.** The instruction is copied from memory
- ○ **C.** The instruction is decoded by the Control Unit
- ○ **D.** The instruction is carried out by the appropriate CPU component

Q5 (1 mark)

What does the Program Counter (PC) register hold?

- ○ **A.** The address of the next instruction to be fetched
- ○ **B.** The number of programs currently running
- ○ **C.** The result of the most recent calculation
- ○ **D.** The current date and time

Q6 (1 mark)

What is the purpose of the Memory Address Register (MAR)?

- ○ **A.** To perform arithmetic calculations
- ○ **B.** To store data being transferred to or from memory
- ○ **C.** To hold the address of the memory location being accessed
- ○ **D.** To store the results of comparisons

Q7 (1 mark)

What is the purpose of the Memory Data Register (MDR)?

- ○ **A.** To hold the address of the next instruction
- ○ **B.** To hold data being transferred between the CPU and memory
- ○ **C.** To store the decoded instruction

- ○ **D.** To control the timing of CPU operations

Q8 (1 mark)

What is the Accumulator used for?

- ○ **A.** To store the address of the current instruction
- ○ **B.** To connect the CPU to the monitor
- ○ **C.** To control which instruction is executed next
- ○ **D.** To store data and the results of ALU operations

Q9 (1 mark)

What is an embedded system?

- ○ **A.** A computer system built into a larger device to perform a dedicated function
- ○ **B.** A computer that is buried underground
- ○ **C.** A type of removable USB storage device
- ○ **D.** A computer that can only browse the internet

Q10 (1 mark)

Which of these is an example of an embedded system?

- ○ **A.** A desktop gaming PC

- ○ **B.** A laptop used for office work
- ○ **C.** A microcontroller in a washing machine controlling the wash cycle
- ○ **D.** A tablet used for watching videos

Q11 (1 mark)

How many times does the FDE cycle repeat while a program runs?

- ○ **A.** Once per program, at the start
- ○ **B.** Only when the user presses a key
- ○ **C.** Ten times per second maximum
- ○ **D.** Once for each instruction in the program

Q12 (1 mark)

What is a key characteristic of embedded systems compared to general-purpose computers?

- ○ **A.** They are always connected to the internet
- ○ **B.** They are designed for a single, specific purpose rather than varied tasks
- ○ **C.** They always have a keyboard and monitor
- ○ **D.** They can only be programmed in machine code

Q13 (2 marks)

Describe the sequence of register operations during the fetch stage.

- ○ **A.** PC → MAR → Memory access → MDR, then PC increments
- ○ **B.** Accumulator → ALU → MDR → MAR
- ○ **C.** ALU → Accumulator → MAR → Memory access
- ○ **D.** MDR → PC → MAR → Accumulator

Q14 (2 marks)

What does the Control Unit do with the instruction during the decode stage?

- ○ **A.** It encrypts the instruction for security purposes
- ○ **B.** It stores the instruction in secondary storage for later use
- ○ **C.** It sends the instruction to the monitor for display
- ○ **D.** It examines the opcode to identify the operation type and prepares control signals to coordinate execution

Q15 (2 marks)

Different instruction types produce different actions during the execute stage.
Which statement correctly matches instruction types to their execute actions?

- ○ **A.** All instructions use the ALU regardless of type

○ **B.** Data transfer instructions modify the Program Counter; branch instructions use the ALU

○ **C.** Arithmetic instructions use the ALU; data transfer instructions move data between registers and memory; branch instructions modify the Program Counter

○ **D.** Branch instructions store data in memory; arithmetic instructions modify the Program Counter

Q16 (2 marks)

At what point in the FDE cycle is the Program Counter typically incremented?

○ **A.** After the instruction is executed, as the final step

○ **B.** During or immediately after the fetch stage, before decode begins

○ **C.** Only when a branch instruction is executed

○ **D.** At the start of the decode stage

Q17 (2 marks)

An embedded system in a car's anti-lock braking system (ABS) monitors wheel speed sensors.
What happens when the system detects a wheel is about to lock?

○ **A.** The system automatically reduces brake pressure to that wheel, preventing lock-up, without requiring driver action

○ **B.** The system displays a warning message and waits for driver input

○ **C.** The system shuts down the engine to stop the car

- ○ **D.** The system sends an email to the car manufacturer

Q18 (2 marks)

Many embedded systems are described as 'real-time' systems. What does this term mean?

- ○ **A.** The system displays the current time on a clock
- ○ **B.** The system only operates during business hours
- ○ **C.** The system must respond to inputs within guaranteed time limits, where late responses are considered failures
- ○ **D.** The system processes data at exactly 1 second intervals

Q19 (2 marks)

Explain how the MAR and MDR work together during a memory read operation.

- ○ **A.** The MDR sends the address to memory, and the MAR receives the data
- ○ **B.** The MAR holds the address of the location to read; memory sends the data at that address to the MDR
- ○ **C.** Both registers hold the same data simultaneously
- ○ **D.** The MAR and MDR only work during write operations, not reads

Q20 (2 marks)

Compare the typical power consumption of an embedded system with a general-purpose PC.

- ○ **A.** Embedded systems typically use more power because they run continuously
- ○ **B.** Power consumption is identical between embedded systems and PCs
- ○ **C.** PCs use no power when in standby mode
- ○ **D.** Embedded systems typically use much less power, often designed for battery operation or minimal heat generation

Q21 (2 marks)

The Current Instruction Register (CIR) holds the instruction currently being processed.
Why is this separate register needed?

- ○ **A.** To free up the MDR for other memory operations while the instruction is being decoded and executed
- ○ **B.** To make the instruction visible on the monitor
- ○ **C.** To allow the user to modify instructions during execution
- ○ **D.** To store multiple instructions at once for parallel processing

Q22 (2 marks)

What factors determine how quickly the FDE cycle can repeat?

- ○ **A.** Only the speed of the keyboard input
- ○ **B.** The CPU clock speed, memory access speed, and the complexity of instructions being executed
- ○ **C.** Only the size of the monitor

- ○ **D.** The number of files stored on the hard drive

Q23 (2 marks)

Why are most embedded systems not designed to be reprogrammed by end users?

- ○ **A.** The manufacturers forgot to include this feature
- ○ **B.** Embedded systems have no memory to store new programs
- ○ **C.** User modification could cause the device to malfunction, and the dedicated function doesn't require software changes
- ○ **D.** Reprogramming embedded systems is illegal in all countries

Q24 (3 marks)

Trace through the complete FDE cycle for a simple ADD instruction that adds the contents of memory location 100 to the Accumulator. Which sequence correctly describes this process?

- ○ **A.** The ADD instruction completes in the decode stage without needing execute
- ○ **B.** Only the Accumulator is used; no memory access is needed
- ○ **C.** Execute happens first, then fetch, then decode
- ○ **D.** Fetch: PC value (say, 50) copied to MAR, instruction retrieved to MDR, PC incremented to 51. Decode: Control Unit identifies ADD instruction and operand address (100). Execute: Address 100 placed in MAR, data retrieved to MDR, ALU adds MDR value to Accumulator, result stored in Accumulator

Q25 (3 marks)

A company is designing an embedded system for a pacemaker (heart rhythm device).
Analyse which constraints are most critical for this application.

- ○ **A.** Reliability, low power consumption, real-time response, and physical size are critical — the device must work flawlessly for years, run on a small battery, respond instantly to heart conditions, and fit inside the human body
- ○ **B.** Maximum processing power is the only consideration
- ○ **C.** Internet connectivity and app store access are essential features
- ○ **D.** Graphics capability and screen resolution are most important

Q26 (3 marks)

The Von Neumann bottleneck affects FDE cycle efficiency.
Analyse how this bottleneck arises from the register and bus architecture.

- ○ **A.** The bottleneck is caused by having too many registers
- ○ **B.** The bottleneck is caused by the Accumulator being too small
- ○ **C.** The bottleneck only affects computers with more than 8 GB of RAM
- ○ **D.** Instructions and data must share the same bus and memory system, so the CPU cannot fetch an instruction and access data simultaneously — when executing an instruction that needs data from memory, the buses are occupied and the next instruction fetch must wait

Q27 (3 marks)

When the CPU encounters a conditional branch instruction (like 'jump if zero'), it doesn't know the outcome until the execute stage.
Analyse why this creates a problem and how modern CPUs address it.

- ○ **A.** There is no problem — the CPU simply waits to see the result before doing anything else
- ○ **B.** The CPU cannot execute branch instructions at all
- ○ **C.** The problem is pipeline stalling: by the time the branch outcome is known, the CPU has already fetched subsequent instructions that may be wrong if the branch is taken; modern CPUs use branch prediction to guess the outcome and speculatively fetch instructions along the predicted path, discarding them if the prediction was wrong
- ○ **D.** Branch instructions are executed before they are decoded

Q28 (3 marks)

Internet of Things (IoT) devices are embedded systems with network connectivity.
Analyse the security challenges this creates compared to traditional isolated embedded systems.

- ○ **A.** IoT devices have no security challenges because they are too simple to hack
- ○ **B.** Network connectivity exposes IoT devices to remote attacks, but their limited resources make implementing strong security difficult; they may have weak/default passwords, unencrypted communications, inability to receive security updates, and become entry points for attacking larger networks
- ○ **C.** IoT devices are more secure than computers because they have no keyboard
- ○ **D.** Security is only a concern for devices that store financial data

Q29 (3 marks)

An interrupt signal arrives while the CPU is in the middle of the FDE cycle.
Analyse how the CPU handles this situation.

- ○ **A.** The CPU completes the current instruction, saves its state (including the PC and register contents) to allow later resumption, then loads the interrupt handler address into the PC to begin handling the interrupt; after handling, the saved state is restored to continue the original program
- ○ **B.** The CPU immediately stops mid-instruction and handles the interrupt
- ○ **C.** Interrupts can only occur when the computer is idle
- ○ **D.** The CPU ignores all interrupts to avoid disrupting the current program

Q30 (3 marks)

Modern cars contain dozens of embedded systems communicating over internal networks.
Analyse why these systems must be carefully isolated and prioritised.

- ○ **A.** All car systems have equal importance and should share resources equally
- ○ **B.** Entertainment systems should have highest priority because passengers use them most
- ○ **C.** Critical safety systems (brakes, steering, airbags) must have guaranteed resources and cannot be disrupted by failures in non-critical systems (entertainment, climate); isolation prevents faults from propagating, and priority ensures safety functions always have the CPU time, memory, and communication bandwidth they need even if other systems fail or misbehave

○ **D.** Modern cars don't actually use embedded systems

Total: **55** marks • Time: **30 min**

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